

SMART RFQ INTELLIGENCE PIPELINE

S R I P

Justification Document

Automating the RFQ-to-Quote Pipeline for Indian Steel MSMEs

The Problem Worth Solving

India's steel distribution and fabrication sector is built on micro, small, and medium enterprises — thousands of traders, stockists, and processors operating out of industrial clusters like Surat, Bhavnagar, and Rajkot. Every day, these businesses receive dozens of Request for Quotation (RFQ) messages — via WhatsApp, email, and paper uploads — each requiring a sales operator to manually parse the requirement, look up steel grades, compute weights using product-specific formulas, fetch live MCX steel prices, apply GST rules, verify BIS/IS standards compliance, and finally type out a structured quote.

A single RFQ can take 20–40 minutes to process manually. For a business handling 30+ RFQs a day, that is an entire dedicated headcount — plus a compounding risk of human error on pricing, weight, or tax calculation.

The Gap in Existing Tools

No existing off-the-shelf tool addresses the end-to-end RFQ workflow for Indian steel SMEs. General-purpose ERP systems are expensive, require weeks of onboarding, and are not designed for the WhatsApp-first communication patterns of Tier-2 and Tier-3 industrial businesses. Spreadsheet-based solutions require manual data entry and are error-prone. There is no product today that can receive a photograph of a handwritten RFQ or a WhatsApp voice-to-text message and return a GST-compliant, BIS-validated, PDF quote within minutes.

Our Target Users

SRIP is built for:

- Inside-sales operators at steel MSMEs who field high-volume RFQs daily across WhatsApp, email, and web uploads
- Small enterprise owners and procurement managers who want faster turnaround on quote generation without hiring additional staff
- Sales teams at steel service centres who need consistent, audit-ready quote documentation with accurate weight and GST breakdowns

Why This Solution

SRIP takes an agent-orchestrated architecture approach — each stage of the quote workflow (OCR, entity extraction, BIS validation, weight computation, live pricing, GST calculation, PDF assembly, and delivery) is handled by a specialised agent. This is not a generic chatbot wrapper. The system encodes real domain knowledge:

- Steel weight formulas per product (TMT bars, plates, pipes, angles, channels, sheets)
- BIS/IS standard codes and material grade validation rules
- HSN-code-based GST classification logic for steel products
- Live MCX steel price lookup via web-search augmentation
- India-specific logistics and freight cost heuristics

The pipeline is designed to ingest RFQs from multiple channels — file upload, raw text, and WhatsApp webhook — making it accessible to businesses regardless of their current workflow. The system gracefully falls back to heuristic logic when AI services are unavailable, ensuring reliability in low-connectivity environments common to Gujarat-tier industrial zones.

Expected Impact and Scalability

Even at conservative estimates, replacing manual RFQ processing for a mid-sized steel trader handling 30 RFQs/day translates to saving 10–15 hours of skilled labour daily, reducing pricing errors, and cutting average quote turnaround from hours to minutes. At scale — across even 1% of Gujarat's 40,000+ registered steel MSMEs — the aggregate productivity gain is significant.

The architecture is cloud-ready: Celery-backed async task processing, Redis queuing, Dockerised deployment, and a ChromaDB knowledge base that can be updated without code changes. New steel product categories, IS codes, or pricing rules can be seeded without redeployment, making the system maintainable by non-engineers.

SRIP addresses a real, underserved operational pain point in the Indian manufacturing supply chain — with a technically credible, domain-specific, and practically deployable solution.